

An Approach To Hyper-Cloud Services Capital and Operational Expenses and Price Estimation

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Revision History

Version	Details
January, 2016	Initial Draft
October, 2016	Opex Calculus
July, 2017	Overload Mgmt. Techniques
November, 2017	Virtual Memory/Disk Capacity Concept

CAPEX / OPEX Baseline

- **CAPEX:**

- New items (1st year)
- ROI as One Time Fee (OTF)

- **OPEX:**

- Renewable items (Starting 2nd year through 1 - 5 years)
- ROI as Recurring Monthly Fee (RMF)
- Includes:
 - Software / Hardware:
 - Copyrights (LTU, License To Use)
 - Life Cycle Support (Upgrade / Replacement)
 - Infrastructure: Facilities (E/S/AC) + Security + Monitoring.
 - Human resources (Operation + Innovation).

OPEX Variation Factors (I)

- **OPEX:**
 - (...)
 - ~~Traditional linear depretiation (3 years).~~
 - **Technology planned obsolescence (TPO)**
 - 3 - 7 years leap.
 - End of sale (New clients).
 - End of support (Warranty / Refurbish).
 - End of life (Operational Risk / Higher costs).

OPEX Variation Factors (II)

- **OPEX:**
 - (...)
 - **Technology innovation gap (TIG)**
 - 1 – 10 years leap.
 - Reduced costs (CAPEX).
 - Increased efficiency (OPEX).

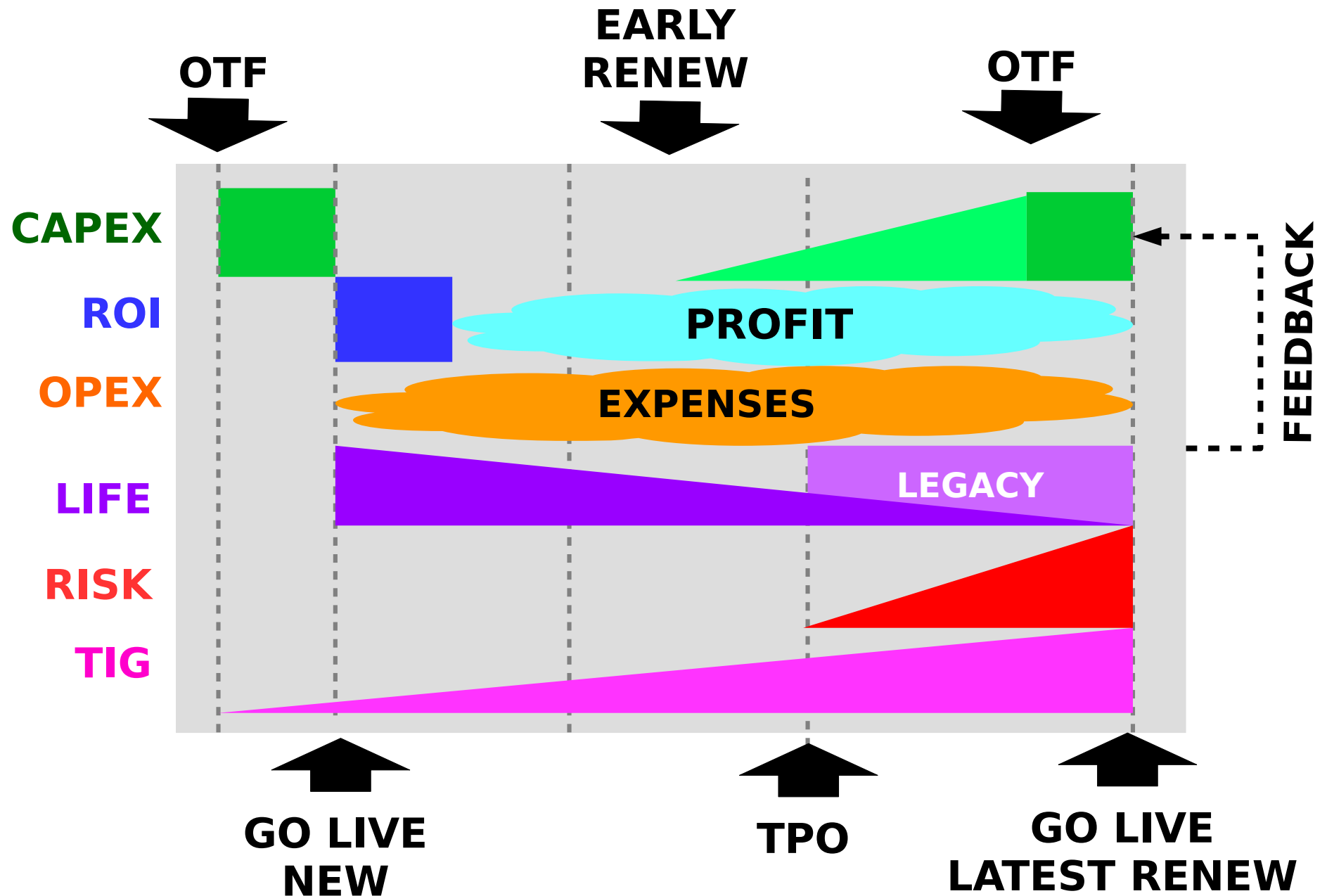
OPEX Variation Factors (III)

- **(Hyper-)Scale economy (HSE)**
 - Capacity acquisition deals (Massive, Forward, New/Old DC).
 - Constant capacity acquisition:
 - Per usage (real capacity)
 - ~~Per sale (provisioned capacity)~~
 - Minimal legacy expenses (LEGEX)
 - Reusing / Mitigation / Innovatio
 - Renovation / Modernization
 - Disposal / Recycling
 - Standarization, Virtualization, Automation & Orchestation.
 - Time as a OPEX increment factor.

Technology Innovation Gap (TIG)

- **Optimization** (Moore's Law):
 - Predictible.
 - **Example:**
 - Higher density disk (HDD).
 - Hardware defined storage (HDS).
- **Modernization** (Kick the table):
 - Unpredictable.
 - **Example:**
 - Solid state disk (SSD).
 - Software defined storage (SDS).

Cost Evolution In Time



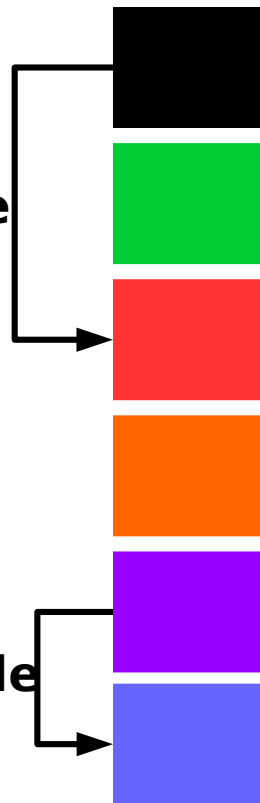
Cloud Structured Cost

- **Install cost.**
 - *On the cloud, Zero install cost. So no OTF.*
- **Service cost.**
 - *On the cloud, a recurring fixed or variable RMF.*
 - **Real Capacity (Installed).**
 - **Virtual Capacity (Oversubscribed)**, depends on the provisioning method:
 - Fixed / No-Oversubscribed / 1:1 (Fixed).
 - Possibility of use / Usage quantity
 - Thin / Oversubscribed / 1:N (Credit).
 - Probability of use / Usage behaviour
 - Architecture grow logic (Vertical / Horizontal).

Cloud Structured Cost



Possible



Real (Installed)

Real Used

Real Reserved

Virtual (Oversubscribed)

Virtual Used

Virtual Protective

Probable

Cloud Structured Cost

- **Is not a Combo!**

- Is a standard... to overcome digital complexity and deliver the solution to more market share.
- ~~Pay for a burger...~~ Pay for a bit...

- **Standard for:**

- **Agile assembly and provisioning** (less handicraft).
- **Enhanced product quality:**
 - Less humans errors and service pitfalls.
 - More service automation opportunities.

Cloud Structured Cost

- **Service cost (Cont.).**
 - (...)
 - **Unused (Idle) Capacity Ratio (UCR):**
 - Protective capacity is not included.
 - Financed by the client / user.
 - Useful for negative demand period.
 - Masquaraded as price discount.
 - **Market (share/complement) behaviour:**
 - Offer.
 - Demand.

Oversubscription

(Overload Problem)

- **Overload occurs when:**
 - Oversubscription is used.
 - Protective idle capacity is consumed.
 - Service degradation or interruption arises.
- **Overload management techniques:**
 - **Stealing resources** (from underloaded VM).
 - **Quiescing** (shutdown / restart VM).
 - **Live migration** (VM workload balancing).
 - **Streaming disks** (partial disk transfer).

Formulas

$$OPEX_{Average} = \frac{\sum_{i=1}^{Months} OPEX_i * (1 + TOG_i + TIG_i) * (1 - HSE_i)}{Months}$$

$0 \leq TOP + TIG < \infty$
 $0 \leq HSE < 1$
 $Months \in \{12, 24, 36, 48, 60\}$

$$CAPACITY_{Virtual} = CAPACITY_{Real} * (1 + OVERSUBSCRIPTION)$$

$0 \leq OVERSUBSCRIPTION < \infty$

$$PRICE_{Average} = \frac{\sum_{i=1}^{Months} OPEX_{Average} * (1 + REVENUE_{Planned}) * (1 - UCR_i)}{Months}$$

$0 \leq REVENUE_{Planned} \leq 0.3$
 $0 < UCR < 0.4$
 $Months \in \{12, 24, 36, 48, 60\}$

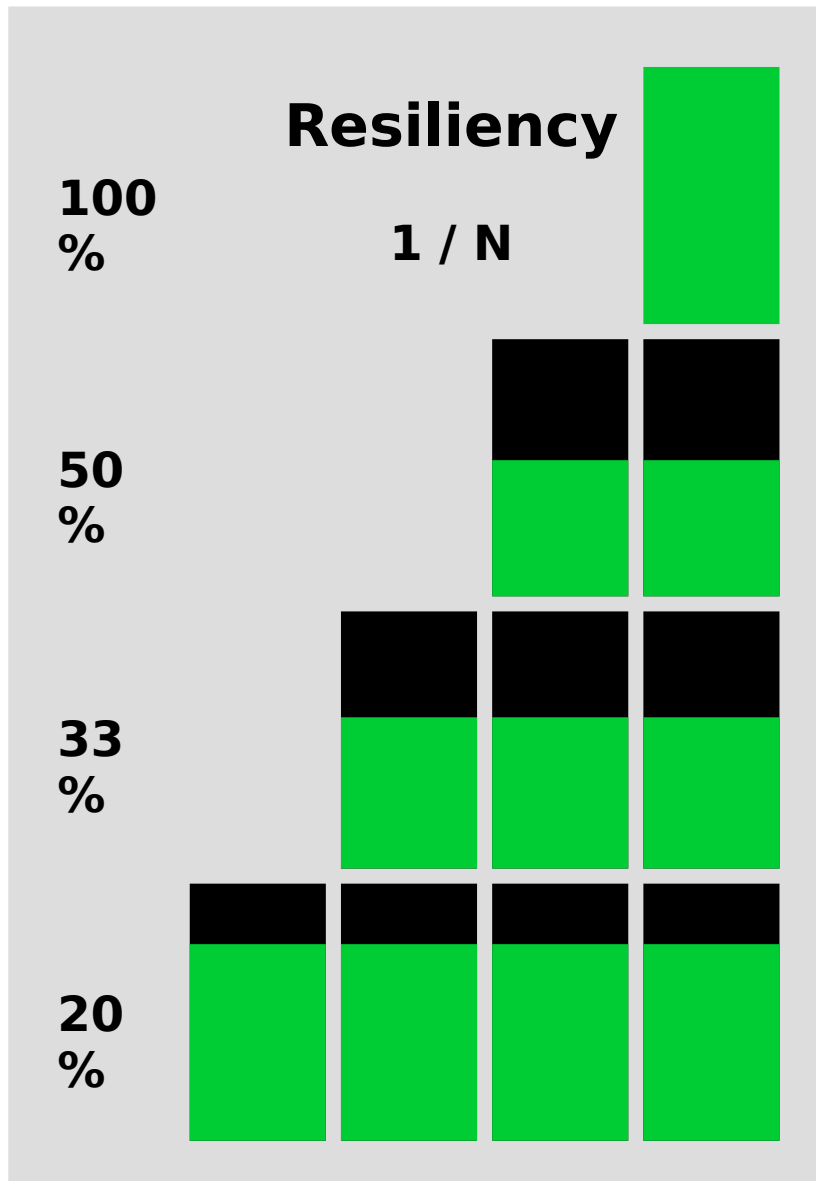
$$REVENUE_{Achieved} = \frac{CAPACITY_{Virtual} * PRICE_{Average}}{CAPACITY_{Real} * OPEX_{Average}}$$

$CAPACITY_{Virtual/Real}, OPEX_{Monthly}, PRICE_{Monthly} > 0$

$$ROI_{Planned} = \frac{CAPEX}{OPEX_{Average}}$$

$$ROI_{Achieved} = \frac{CAPEX}{PRICE_{Average}}$$

Virtual Memory Capacity

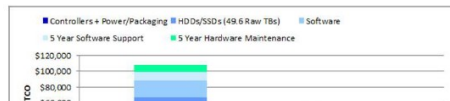


- **Over-Subscription:**
 - From 1:1 to 1:N (Thin Prov.)
 - Overload risks & automation.
 - Horizontal scale & logistics.
- **Heterogeneous Farms Consolidation:**
 - VMWare CPU policy restrictions.

Virtual Disk Capacity

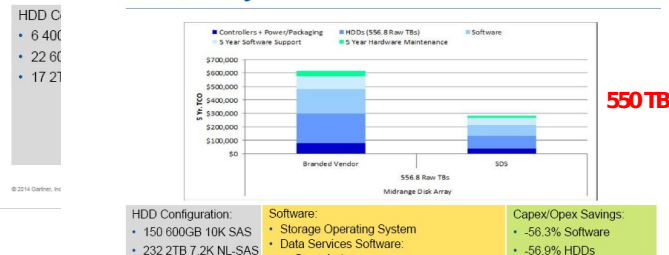
2014

Analysis of a Small Hybrid Disk Array



2014

Analysis of a Typical Midrange Disk Array



• Over-Subscription:

- From 1:1 to 1:N (Thin Prov.).
- Overload risks & automation.
- Horizontal scale & logistics.

• Technology Innovation Gap:

- **Hardware Defined Storage**
 - EMC CX*, VNX*, FC SAN.
- **Software Defined Storage**
 - Server, ETH SAN.

Solution	Type	5-year TCO
Dell EMC Unity 300	Disk Array	\$330,865
Red Hat Enterprise Storage (Ceph)	Software Defined Storage	\$70,047
VMware Virtual SAN 6		
DataCore SANsymphony		
NetApp FAS2554		
IBM v5010		
Scality Ring		
SUSE Enterprise Storage 4		

Capex/Opex Savings:

- -56.3% Software
- -56.9% HDDs
- -53.1% Controller + P&P
- -54.1% Net Price per GB
- \$332,569 5-Year Savings

>50% Av. Saving

Gartner.

TIG ≥ 0

